Project 1 based on Angrist and Krueger (1991, QJE)

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Project 1

In this Project we will try and replicate the different graphs and tables

I.Preliminaries

```
library(tidyverse) #we will be using tidyverse to be able to graph using ggplot
library(here)
library(lubridate)
library(gridExtra)
library(stargazer) #stargazer will be used in Question 2 to export tables
library(AER)        ## for ivreg, this will be used to make the IVregressions
library(ipumsr)     ##for extracting ipums data
library(dplyr)
library(kableExtra)  ##to make tables
```

We used the code written in the instructions of the project to be able to set the correct data base named pums.

```
pums <- read.table(paste("QOB.raw",sep=""),
                   header           = FALSE,
                   stringsAsFactors = FALSE)
colnames(pums)[c(1,2,4,5,6,9:13,16,18,19:21,24,25,27)] <- c("AGE", "AGEQ", "EDUC",
                                                            "ENOCENT","ESOCENT", "LWKLYWGE",
                                                            "MARRIED", "MIDATL", "MT", "NEWEN
G", "CENSUS", "QOB", "RACE",
                                                            "SMSA", "SOATL", "WNOCENT", "WSOC
ENT", "YOB")
pums <- as_tibble(pums)

pums %>%
  mutate(cohort = factor(1*(YOB<=39 & YOB >=30) +
                           2*(YOB<=49 & YOB >=40),
                         levels=c(1,2), labels=c("30-39","40-49")) ) -> pums

ymd(paste("19",30,4 * 3, sep=""),truncated = 2)
```

```
## [1] "1930-12-01"
```

```
pums <- as_tibble(pums)
```

II.Question 1

Explain the research design of the paper (1). In particular, explain why estimation by OLS is likely to be invalid (2), explain the nature of the instruments (3), and discuss whether the proposed instrumental variable strategy is credible. (4)

1. The research design of the paper is to determine the causal relationship between education and earnings using instrumental variable. The reason of this choice is due to fix some endogeneity problems in ordinary least squares regression.

2. OLS estimates may be biased and inconsistent due to endogeneity issues:

- Some omitted factors like motivation might affect both education and earnings. That's the main reason why correlation doesn't imply causality. (omitted variables)
- Higher earnings may also get individuals to continue more years of studying. (reverse effect)
- Errors in measuring years of schooling make some bias in OLS estimates. (errors measurement)

3. The authors use 'quarter of birth' as an instrumental variable for years of schooling. The authors suggest that individuals may have different levels of schooling because of the timing of attending class according the months of their birth and when they leave school.

4. 2 factors can contribute to the credibility of the instrumental variable strategy.

- First, the instrument must be correlated with the endogenous variable (education).
- Secondly the instrument must not be directly related to the outcome variable (earnings), only via its effect on education.

III.1.1 Question 2

In this Question we have been tasked to replicate the graph 1 which is a graph showing the evolution between the Year of Birth and their Educational Levels, and we will have to graph the graph 5 which is the evolution between the Year of Birth and the wage (in log average earnings).

These are the steps to be able to replicate the graphs :

Step 1 : Changing the date format so that we can have clearer values to show in the graph.

```
pums <- pums %>%
  mutate(YOB = as.Date(paste("19", YOB, "01-02", sep = ""), format = "%Y%d-%m"))
```

Step 2 : We then calculate the average of EDUC and LWKLYWGE to have one point per year of birth.

```
educ_mean <- pums %>%
  group_by(YOB, QOB) %>%
  summarise(mean_EDUC = mean(EDUC),
            mean_LWKLYWGE = mean(LWKLYWGE),
            .groups = 'keep')
```

Step 3 : Creation of a variable associating year and quarter of birth

```
educ_mean <- educ_mean %>%
  mutate(YQOB = paste(YOB, QOB, sep = "-"))
```

Step 4 : We plot the first figure into the object p1

```
#Creation of 2 graphics
p1 <-ggplot(educ_mean, aes(x = YQOB, y = mean_EDUC, label = as.character(QOB), color = as.cha
racter(QOB))) +
  geom_point(size = 3, shape = 16) +
  geom_line(aes(group = 1), color = "black") +
  geom_text(aes(label = QOB), color = "white", size = 2) +
  scale_color_manual(values = c("1" = "red", "2" = "black", "3" = "black", "4" = "black"), gu
ide = "none") +
  labs(x = "Year of birth", y = "Years of completed education", title = "Figure 1 : Average e
ducation by quarter of birth") +
  scale_x_discrete(breaks = c("1930-02-01-1","1935-02-01-1", "1940-02-01-1",
                              "1945-02-01-1", "1949-02-01-4"),
                   labels = c("1930-02-01-1" = "1930","1935-02-01-1" ="1935",
                              "1940-02-01-1" = "1940","1945-02-01-1" = "1945",
                              "1949-02-01-4" = "1950"))
```

Step 5 : We then plot the second figure into the object p2

```
p2 <-ggplot(educ_mean, aes(x = YQOB, y = mean_LWKLYWGE, label = as.character(QOB), color = a
s.character(QOB))) +
  geom_point(size = 3, shape = 16) +
  geom_line(aes(group = 1), color = "black") +
  geom_text(aes(label = QOB), color = "white", size = 2) +
  scale_color_manual(values = c("1" = "red", "2" = "black", "3" = "black", "4" = "black"), gu
ide = "none") +
  labs(x = "Year of birth", y = "Log weekly earnings", title = "Figure 2 : Average weekly wag
e by quarter of birth") +
  scale_x_discrete(breaks = c("1930-02-01-1","1935-02-01-1",
                              "1940-02-01-1","1945-02-01-1", "1949-02-01-4"),
                   labels = c("1930-02-01-1" = "1930","1935-02-01-1" ="1935",
                              "1940-02-01-1" = "1940","1945-02-01-1" = "1945",
                              "1949-02-01-4" = "1950"))
```

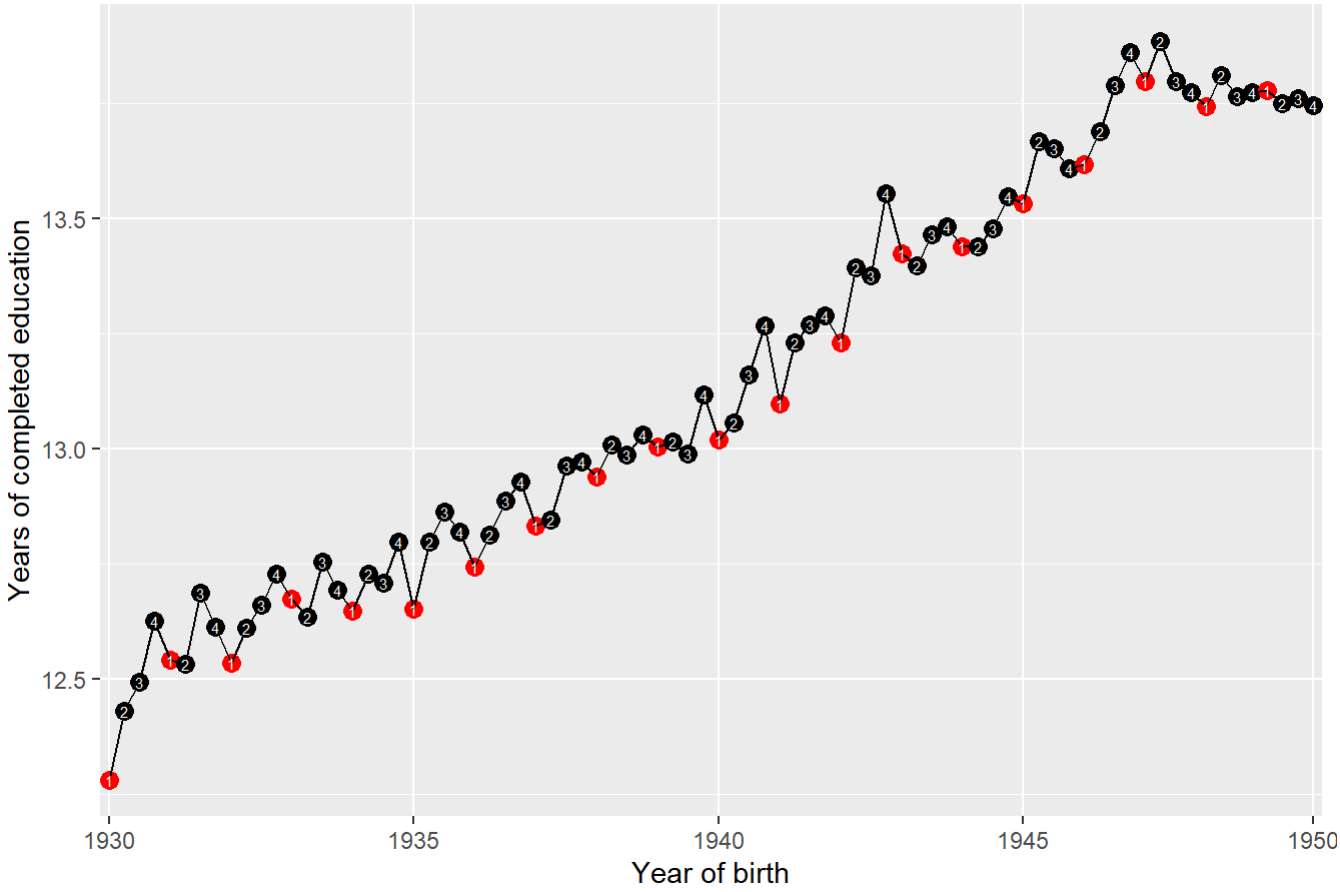
III.1.2 Figure 1

We then View our graphs.

On this figure, we can observe the year of education reached by quarter of birth for each year between 1930 and 1950. Overall, there is an increase in years of education independently of the quarter of birth with a beginning of a stagnation since the middle of 1947 to 1950. The first quarter of birth is always the one with the least years of education for each year.

```
p1
```

Figure 1 : Average education by quarter of birth

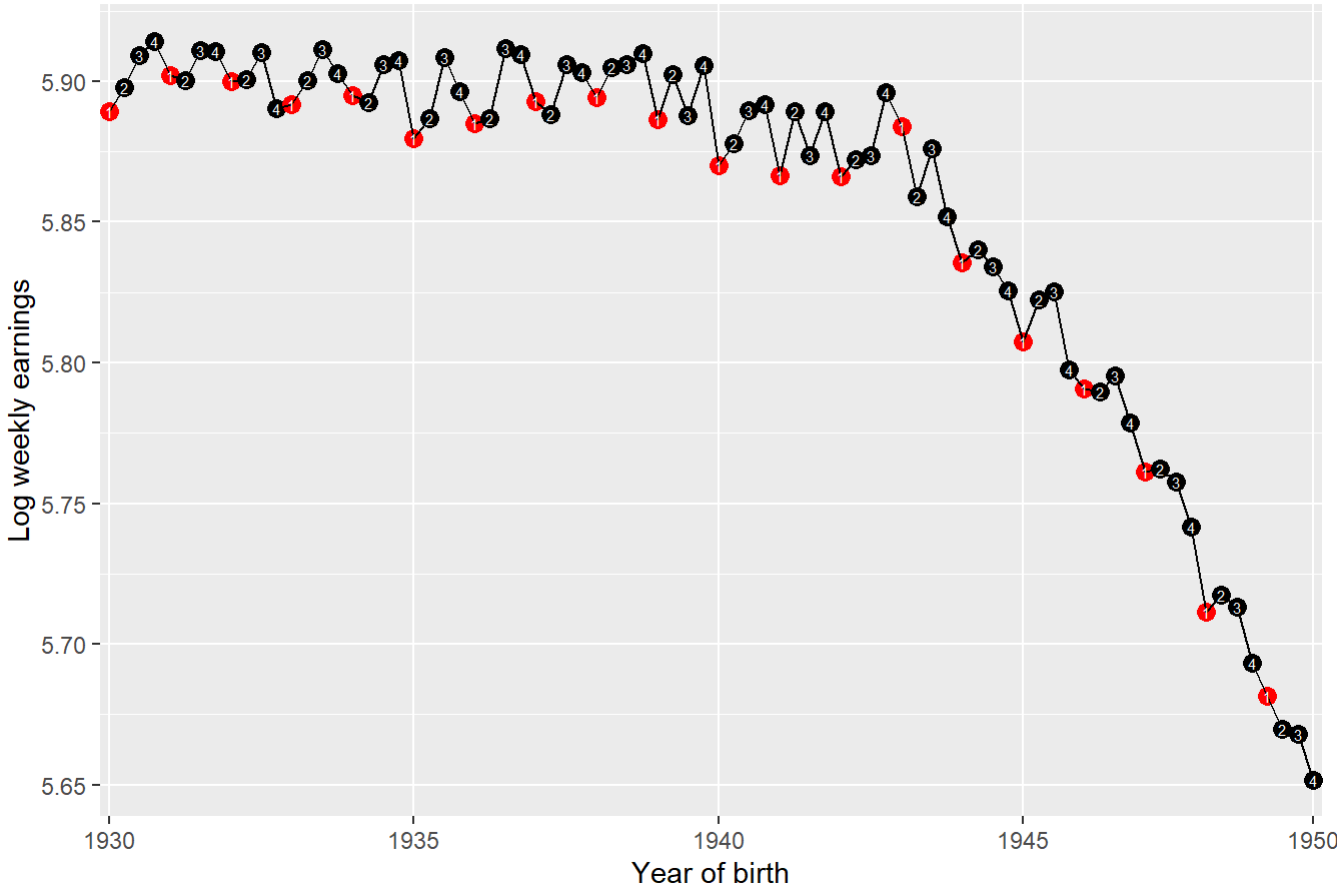


III.1.3 Figure 2

On this figure, we can observe the wage (in log average earnings) reached by quarter of birth for each year between 1930 and 1950. Overall, there is a decrease in wage independently of the quarter of birth which begins in 1942. Before this point, there is a stagnation. Surprisingly, during the stagnation phase, the first quarter of birth is the one with the least wage but after 1942, when the slope becomes more abrupt, it's the one with the highest wage per week.

```
p2
```

Figure 2 : Average weekly wage by quarter of birth



III.1.4 Wald Estimator

In Question 2 we have been tasked to make the table Wald Estimator. We must first understand what it is is : *The Wald estimator determines the return to education by dividing the difference in earnings per quarter of birth by the difference in years of education per quarter of birth. Here we compare income and years of education between men born in the first trimester and those born in the second, third and fourth quarters, because men in the first trimester are on average the most educated and have the highest salaries.*

```
pums %>%
    filter(cohort == "30-39") -> pums.tab5

pums.tab5$wald_dum <- (pums.tab5$QOB == 3 | pums.tab5$QOB == 4) * 1
```

The following code is the method to extract the Wald Estimator:

Step 1 : Mutating pums.tab5 to define the variable wald_dum

```
pums.tab5 <- pums.tab5 %>%
  mutate(wald_dum = as.numeric(QOB == 2 | QOB == 3 | QOB == 4))
```

Step 2 : Calculate the means for the groups

```
mean_ln_wage_q1 <- mean(pums.tab5$LWKLYWGE[pums.tab5$QOB == 1], na.rm = TRUE)
mean_ln_wage_q2_4 <- mean(pums.tab5$LWKLYWGE[pums.tab5$wald_dum == 1], na.rm = TRUE)
mean_education_q1 <- mean(pums.tab5$EDUC[pums.tab5$QOB == 1], na.rm = TRUE)
mean_education_q2_4 <- mean(pums.tab5$EDUC[pums.tab5$wald_dum == 1], na.rm = TRUE)
```

Step 3 : Calculate the Wald estimator

```
wald_estimator <- (mean_ln_wage_q1 - mean_ln_wage_q2_4) / (mean_education_q1 - mean_education
_q2_4)
```

Step 4 : Calculate the group sizes

```
n_q1 <- sum(!is.na(pums.tab5$LWKLYWGE[pums.tab5$QOB == 1]))
n_q2_4 <- sum(!is.na(pums.tab5$LWKLYWGE[pums.tab5$wald_dum == 1]))
```

Step 5 : Calculate the variances

```
var_ln_wage_q1 <- var(pums.tab5$LWKLYWGE[pums.tab5$QOB == 1], na.rm = TRUE)
var_ln_wage_q2_4 <- var(pums.tab5$LWKLYWGE[pums.tab5$wald_dum == 1], na.rm = TRUE)
var_education_q1 <- var(pums.tab5$EDUC[pums.tab5$QOB == 1], na.rm = TRUE)
var_education_q2_4 <- var(pums.tab5$EDUC[pums.tab5$wald_dum == 1], na.rm = TRUE)
```

Step 6 : Calculate the standard errors for the differences

```
se_ln_wage_diff <- sqrt(var_ln_wage_q1 / n_q1 + var_ln_wage_q2_4 / n_q2_4)
se_education_diff <- sqrt(var_education_q1 / n_q1 + var_education_q2_4 / n_q2_4)
```

Step 7 : Calculate the standard error for the Wald estimator

```
wald_se <- sqrt(
  (se_ln_wage_diff^2 / (mean_education_q1 - mean_education_q2_4)^2) +
    ((mean_ln_wage_q1 - mean_ln_wage_q2_4)^2 * se_education_diff^2 / (mean_education_q1 - mea
n_education_q2_4)^4)
)
```

Step 8 : Adjust the OLS regression for the return to education

```
reg_ols <- lm(LWKLYWGE ~ EDUC, data = pums.tab5)
```

Step 9 : Retrieve the OLS results

```
ols_return <- coef(reg_ols)["EDUC"]
ols_se <- summary(reg_ols)$coefficients["EDUC", 2]
```

Step 10 : Create the results table with standard errors

```
data3 <- data.frame(
  ln_wkly_wage = c(
    round(mean_ln_wage_q1, 5),
    round(mean_ln_wage_q2_4, 5),
    paste0(round(mean_ln_wage_q1 - mean_ln_wage_q2_4, 5), " (", round(se_ln_wage_diff, 5),
")")
  ),
  Education = c(
    round(mean_education_q1, 4),
    round(mean_education_q2_4, 4),
    paste0(round(mean_education_q1 - mean_education_q2_4, 4), " (", round(se_education_diff,
4), ")")
  ),
  Wald_estimator = c(
    "", "",
    paste0(round(wald_estimator, 4), " (", round(wald_se, 4), ")")
  ),
  OLS_return_of_education = c(
    "", "",
    paste0(round(ols_return, 4), " (", round(ols_se, 4), ")")
  ),
  row.names = c(
    "Born in 1st quarter of year",
    "Born in 2nd, 3rd or 4th quarter of year",
    "Difference (std. error) (1)-(2)"
  )
)

# Transpose the data for flipping
data3_flipped <- as.data.frame(t(data3))

# Fix row and column names for better readability
rownames(data3_flipped) <- c(
  "ln (wky. wage)",
  "Education",
  "Wald est. of return to education",
  "OLS return to education"
)
colnames(data3_flipped) <- c("(1)", "(2)", "(3)")

# Generate the flipped table
t1 <- kable(data3_flipped, format = "html", escape = FALSE) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  add_header_above(c(" " = 1, "Born in 1st quarter of year" = 1, "Born in 2nd, 3rd, or 4th qu
arter of year" = 1, "Difference (std. error)" = 1))

t1
```

	Born in 1st quarter of year	Born in 2nd, 3rd, or 4th quarter of year	Difference (std. error)
	1.	2.	3.
ln (wky. wage)	5.8916	5.90269	-0.0111 (0.00274)

	Born in 1st quarter of year	Born in 2nd, 3rd, or 4th quarter of year	Difference (std. error)
	1.	2.	3.
Education	12.6881	12.7969	-0.1088 (0.0133)
Wald est. of return to education			0.102 (0.0281)
OLS return to education			0.0709 (3e-04)

The Wald estimator of the return to education shows that an additional year of education is linked to an average increase of 10.2% in weekly salary (which is in logarithm) for individuals influenced by the instrument, here the quarter of birth. This context highlights that the Wald estimator evaluates the local causal effect for those whose educational level is affected by quarter of birth (QOB). The standard deviation of 0.0239 indicates the uncertainty surrounding this estimate, meaning that the estimated causal effect is statistically significant, as 0.1020/0.0239 ≈ 4.27, resulting in a high t-score (greater than 2) for a threshold of 5%. Regarding the return to education according to the OLS, this estimate reveals that an additional year of education is associated with an average increase of 7.09% in weekly wages, when calculated by ordinary regression (OLS). However, OLS may be biased due to endogeneity: unobserved factors, such as ability or motivation, may influence both education and salary. This could explain why the OLS estimate is lower than that of the Wald estimator. The very low standard deviation (0.0003) indicates high statistical precision of the OLS, but this precision can be misleading if the estimate is biased. And that is what we can see when we know that QOB is somewhat linked to the years of educations.

The table shows that men born in the 1st quarter have slightly less education (-0.11 years) and lower earnings (-1.1%) than those born later. The Wald estimate of returns to education is 10.2%, higher than the OLS estimate 7.1%), indicating higher returns when accounting for biases.

III.1.5 Table 5 (OLS and IV)

We will be doing the code of Table 5 which implies to make regression models.

Step 1 : We mutate pums for years from 1930 to 1939 and turn it into pums.tab5

```
pums %>%
  filter(cohort == "30-39") -> pums.tab5
```

Step 2: We create the region variable by combining the regional variables

```
pums.tab5 %>%
  mutate(region = as.factor(paste(NEWENG, MIDATL, ENOCENT, WNOCENT, SOATL, ESOCENT, WSOCENT,
MT, sep = "_"))) -> pums.tab6
```

Step 3: We create the YOB_factor for Year of Birth fixed effects

```
pums.tab6 %>%
  mutate(YOB_factor = as.factor(YOB)) -> pums.tab6
```

Step 4: Then we create the OLS Regressions

```
reg1 = lm(LWKLYWGE ~ EDUC + YOB_factor, data = pums.tab6)
reg3 = lm(LWKLYWGE ~ EDUC + YOB_factor + AGE + I(AGE^2),pums.tab6)
reg5 = lm(LWKLYWGE ~ EDUC + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_factor + regi
on, data = pums.tab6)
reg7 = lm(LWKLYWGE ~ EDUC + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_factor + AGE
+ I(AGE^2) + region, data = pums.tab6)

is_dummy_included <- function(model, variables) {
  included_vars <- names(coef(model))
  any(grepl(paste(variables, collapse = "|"), included_vars))
}
```

Step 5 : Dummy Variables for Detection

```
year_dummies <- paste0("YOB_factor", levels(pums.tab6$YOB_factor))
region_dummies <- levels(pums.tab6$region)
```

Step 6 : Yes/No Helper Function

```
yes_no <- function(logical_value) {
  if (logical_value) "Yes" else "No"
}
```

Step 7: Dynamic Add Lines

```
ols_year_dummies <- sapply(list(reg1, reg3, reg5, reg7), function(model) is_dummy_included(mo
del, year_dummies))
ols_region_dummies <- sapply(list(reg1, reg3, reg5, reg7), function(model) is_dummy_included
(model, region_dummies))
```

Step 8: Creating the OLS Summary Table

```
stargazer(
  reg1, reg3,reg5, reg7,
  dep.var.caption = "",
  dep.var.labels = "",
  omit = c("YOB_factor", "region", "Constant"), # omit the fixed effects for year of birth an
d region
  add.lines = list(
    c("9 Year-of-birth dummies", sapply(ols_year_dummies, yes_no)),
    c("8 Region-of-residence dummies", sapply(ols_region_dummies, yes_no))
  ),
  star.cutoffs = NA,
  keep.stat = c("n", "rsq"),
  no.space = TRUE,
  header = FALSE,
  column.labels = c("OLS", "OLS", "OLS","OLS"),
  title = "OLS Estimation of the Return to Education for Men Born 1930-1939: 1980 Census",
  type = "text"
)
```

```
##
## OLS Estimation of the Return to Education for Men Born 1930-1939: 1980 Census
## ================================================================
##
##                                 OLS      OLS      OLS      OLS
##                                 (1)      (2)      (3)      (4)
## ----------------------------------------------------------------
## EDUC                          0.071    0.071    0.063    0.063
##                              (0.0003) (0.0003) (0.0003) (0.0003)
## factor(RACE)1                                   -0.257   -0.257
##                                                (0.004)  (0.004)
## factor(SMSA)1                                   -0.176   -0.176
##                                                (0.003)  (0.003)
## factor(MARRIED)1                                 0.248    0.248
##                                                (0.003)  (0.003)
## AGE                                    -0.048            -0.042
##                                       (0.040)           (0.039)
## I(AGE2)                                 0.0005            0.0004
##                                       (0.0004)          (0.0004)
## ----------------------------------------------------------------
## 9 Year-of-birth dummies         Yes      Yes      Yes      Yes
## 8 Region-of-residence dummies    No       No      Yes      Yes
## Observations                  329,509  329,509  329,509  329,509
## R2                             0.118    0.118    0.165    0.165
## ================================================================
## Note:                                                       NA
```

Step 9: Creating QOB_Year Interaction

```
pums.tab6 <- pums.tab6 %>%
  mutate(QOB_Year = paste(QOB, YOB, sep = "_"))
```

Step 10: TSLS Regressions (Instrumental Variable Regressions)

```
reg_IV2 <- ivreg(LWKLYWGE ~ EDUC + YOB_factor | QOB_Year + YOB_factor, data = pums.tab6)
reg_IV4 <- ivreg(LWKLYWGE ~ EDUC  + YOB_factor + AGE +  I(AGE^2)  |
                   QOB_Year  + YOB_factor+ AGE +  I(AGE^2), data = pums.tab6)
reg_IV6 <- ivreg(LWKLYWGE ~ EDUC + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_factor
+ region | QOB_Year + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_factor + region, da
ta = pums.tab6)
reg_IV8 <- ivreg(LWKLYWGE ~ EDUC + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_factor
+ AGE +  I(AGE^2) + region | QOB_Year + factor(RACE) + factor(SMSA) + factor(MARRIED) + YOB_f
actor + region, data = pums.tab6)

tsls_year_dummies <- sapply(list(reg_IV2, reg_IV4, reg_IV6, reg_IV8), function(model) is_dumm
y_included(model, year_dummies))
tsls_region_dummies <- sapply(list(reg_IV2, reg_IV4, reg_IV6, reg_IV8), function(model) is_du
mmy_included(model, region_dummies))
```

Step 11 : Calculating the statistics χ^2

```r
compute_chisq <- function(reg_full, reg_restricted) {
  # Calculating of the Residues
  rss_full <- sum(resid(reg_full)^2)
  rss_restricted <- sum(resid(reg_restricted)^2)

  # Calculation of the degrees of freedom
  df_full <- length(coef(reg_full)) - 1  # get rid of the intercept
  df_restricted <- length(coef(reg_restricted)) - 1

  # Formula for \( \chi^2 \)
  chisq_stat <- (rss_restricted - rss_full) / rss_full * df_full

  # Return \( \chi^2 \) and the degrees of freedom
  return(list(chisq = chisq_stat, df = df_full - df_restricted))
}

# Restraints models for each Tsls
restricted_reg_IV2 <- lm(LWKLYWGE ~ 1, data = pums.tab6)  # Nul Model
restricted_reg_IV4 <- lm(LWKLYWGE ~ 1, data = pums.tab6)
restricted_reg_IV6 <- lm(LWKLYWGE ~ 1, data = pums.tab6)
restricted_reg_IV8 <- lm(LWKLYWGE ~ 1, data = pums.tab6)

# Calculation of the\( \chi^2 \)
chisq_IV2 <- compute_chisq(reg_IV2, restricted_reg_IV2)
chisq_IV4 <- compute_chisq(reg_IV4, restricted_reg_IV4)
chisq_IV6 <- compute_chisq(reg_IV6, restricted_reg_IV6)
chisq_IV8 <- compute_chisq(reg_IV8, restricted_reg_IV8)
```

Step 12 : TSLS table

```
stargazer(
  reg_IV2, reg_IV4,  reg_IV6, reg_IV8,
  dep.var.caption = "",
  dep.var.labels = "",
  omit = c("YOB_factor", "region", "Constant"),
  add.lines = list(
    c("9 Year-of-birth dummies", sapply(tsls_year_dummies, yes_no)),
    c("8 Region-of-residence dummies", sapply(tsls_region_dummies, yes_no)),
    c("\u03C7\u00B2 (dof)",              # Row for \( \chi^2 \)
      paste0(round(chisq_IV2$chisq, 2), " [", chisq_IV2$df, "]"), # TSLS Column 2
      paste0(round(chisq_IV4$chisq, 2), " [", chisq_IV4$df, "]"), # TSLS Column 4
      paste0(round(chisq_IV6$chisq, 2), " [", chisq_IV6$df, "]"), # TSLS Column 6
      paste0(round(chisq_IV8$chisq, 2), " [", chisq_IV8$df, "]")  # TSLS Column 8
    )
  ),
  star.cutoffs = NA,
  keep.stat = c("n", "rsq"), # show only n and R2
  no.space = TRUE,
  header = FALSE,
  column.labels = c("TSLS", "TSLS", "TSLS", "TSLS"),
  title = " TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR MEN BORN 1930-1939: 1980 CENSUS",
  type = "text",
  covariate.labels = c(
    "Years of education",
    "Race (1 = Black)",
    "SMSA (1 = center city)",
    "Married (1 = married)",
    "Age",
    "Age-squared"
  ),
  digits = 4
)
```

```
##
## TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR MEN BORN 1930-1939: 1980 CENSUS
## ============================================================================
##
##                                 TSLS      TSLS      TSLS      TSLS
##                                  (1)       (2)       (3)       (4)
## ----------------------------------------------------------------------------
## Years of education             0.0891    0.0832    0.0806    0.0766
##                               (0.0161)  (0.0221)  (0.0164)  (0.0219)
## Race (1 = Black)                                   -0.2302   -0.2364
##                                                   (0.0261)  (0.0348)
## SMSA (1 = center city)                             -0.1581   -0.1622
##                                                   (0.0174)  (0.0232)
## Married (1 = married)                              0.2440    0.2449
##                                                   (0.0049)  (0.0059)
## Age                                      -0.0517             -0.0467
##                                         (0.0407)            (0.0396)
## Age-squared                              0.0005              0.0005
##                                         (0.0005)            (0.0004)
## ----------------------------------------------------------------------------
## 9 Year-of-birth dummies          Yes       Yes       Yes       Yes
## 8 Region-of-residence dummies    No        No        Yes       Yes
## χ² (dof)                      1.24 [10] 1.55 [12] 3.95 [21] 4.42 [23]
## Observations                   329,509   329,509   329,509   329,509
## R2                             0.1102    0.1143    0.1584    0.1611
## ============================================================================
## Note:                                                                   NA
```

Step 13 : Making the combined table with the ols regressions and the Tsls

```
stargazer(
  reg1, reg_IV2, reg3, reg_IV4, reg5, reg_IV6, reg7, reg_IV8,
  dep.var.caption = "",
  dep.var.labels = "",
  omit = c("YOB_factor", "region", "Constant"),
  add.lines = list(
    c("9 Year-of-birth dummies",
      yes_no(ols_year_dummies[1]), yes_no(tsls_year_dummies[1]),
      yes_no(ols_year_dummies[2]), yes_no(tsls_year_dummies[2]),
      yes_no(ols_year_dummies[3]), yes_no(tsls_year_dummies[3]),
      yes_no(ols_year_dummies[4]), yes_no(tsls_year_dummies[4])
    ),
    c("8 Region-of-residence dummies",
      yes_no(ols_region_dummies[1]), yes_no(tsls_region_dummies[1]),
      yes_no(ols_region_dummies[2]), yes_no(tsls_region_dummies[2]),
      yes_no(ols_region_dummies[3]), yes_no(tsls_region_dummies[3]),
      yes_no(ols_region_dummies[4]), yes_no(tsls_region_dummies[4])
    ),
    c("\u03C7\u00B2 (dof)",              # Lines for \( \chi^2 \)
      "",                               # OLS Column (empty for \( \chi^2 \))
      paste0(round(chisq_IV2$chisq, 2), " [", chisq_IV2$df, "]"), # TSLS column 2
      "",                               # OLS Column (empty for \( \chi^2 \))
      paste0(round(chisq_IV4$chisq, 2), " [", chisq_IV4$df, "]"), # TSLS column 4
      "",                               # OLS Column (empty for \( \chi^2 \))
      paste0(round(chisq_IV6$chisq, 2), " [", chisq_IV6$df, "]"), # TSLS column 6
      "",                               # OLS Column (empty for \( \chi^2 \))
      paste0(round(chisq_IV8$chisq, 2), " [", chisq_IV8$df, "]")  # TSLS column 8
    )
  ),
  star.cutoffs = NA,
  keep.stat = c("n", "rsq"), # Show only n and R^2
  no.space = TRUE,
  header = TRUE,
  column.labels = c("OLS", "TSLS", "OLS", "TSLS", "OLS", "TSLS", "OLS", "TSLS"),
  title = "OLS AND TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR MEN BORN 1930-1939: 1980 CEN
SUS",
  type = "text",
  covariate.labels = c(
    "Years of education",
    "Race (1 = Black)",
    "SMSA (1 = center city)",
    "Married (1 = married)",
    "Age",
    "Age-squared"
  ),
  digits = 4
)
```

```
##
## OLS AND TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR MEN BORN 1930-1939: 1980 CENSUS
## ================================================================================================
============================
##
##                                 OLS    instrumental   OLS    instrumental   OLS    instrum
ental   OLS    instrumental
##                                          variable             variable             varia
ble              variable
##                                 OLS        TSLS       OLS        TSLS       OLS        TSL
S       OLS        TSLS
##                                 (1)        (2)        (3)        (4)        (5)        (6)
(7)        (8)
## ------------------------------------------------------------------------------------------
--------------------------
## Years of education            0.0711     0.0891     0.0711     0.0832     0.0632     0.08
06     0.0632     0.0766
##                                (0.0003)   (0.0161)   (0.0003)   (0.0221)   (0.0003)   (0.01
64)   (0.0003)   (0.0219)
## Race (1 = Black)                                                            -0.2575    -0.23
02    -0.2575    -0.2364
##                                                                             (0.0040)   (0.02
61)   (0.0040)   (0.0348)
## SMSA (1 = center city)                                                      -0.1763    -0.15
81    -0.1763    -0.1622
##                                                                             (0.0029)   (0.01
74)   (0.0029)   (0.0232)
## Married (1 = married)                                                       0.2479     0.24
40     0.2479     0.2449
##                                                                             (0.0032)   (0.00
49)   (0.0032)   (0.0059)
## Age                                                 -0.0475    -0.0517
-0.0421    -0.0467
##                                                     (0.0399)   (0.0407)
(0.0388)   (0.0396)
## Age-squared                                          0.0005     0.0005
0.0004     0.0005
##                                                     (0.0004)   (0.0005)
(0.0004)   (0.0004)
## ------------------------------------------------------------------------------------------
--------------------------
## 9 Year-of-birth dummies        Yes        Yes        Yes        Yes        Yes        Yes
Yes        Yes
## 8 Region-of-residence dummies    No         No         No         No        Yes        Yes
Yes        Yes
## χ² (dof)                                 1.24 [10]            1.55 [12]            3.95
[21]            4.42 [23]
## Observations                  329,509    329,509    329,509    329,509    329,509    329,5
09    329,509    329,509
## R2                             0.1177     0.1102     0.1178     0.1143     0.1650     0.15
84     0.1650     0.1611
## ================================================================================================
============================
## Note:
NA
```

In the tables, we do not have the correct values for the khi². This may be due to not having the same data base. If we had the original data base we may be able to extract the same values needed for the khi².

The table shows OLS and TSLS estimates of returns to education using 1980 Census data. OLS estimates a 7.1% increase in earnings per year of education, while TSLS suggests a higher return 8.9% but with more variability. More variables (like race, marital status) show significant effects on earnings. TSLS addresses potential bias but has larger standard errors, questioning instrument reliability.

IV.Question 3

To be able to extract ipums US 1980 we need to use the code used in Homework 5. We must extract the necessary variables to replicate what we did in Question 2 which are Birth year, level of education, race, birth quarter, and income wage :

```
# the ACS 1980 has id: us1980d
usa_ext_def <- define_extract_usa(
  description = "Extract for individuals born between 1930 and 1939",
  samples = c("us1980a"),
  variables = c("BIRTHYR", "EDUC", "AGE", "RACE", "BIRTHQTR", "INCWAGE")
)

usa_ext_submitted <- submit_extract(usa_ext_def)

usa_ext_complete <- wait_for_extract(usa_ext_submitted)

filepath <- download_extract(usa_ext_submitted,
                             download_dir = "~")

ddi <- read_ipums_ddi(filepath)
dat1980 <- read_ipums_micro(ddi)

# Convert column names to lowercase
colnames(dat1980) <- str_to_lower(colnames(dat1980))
```

To compare the QQB.raw data from the 1980census Ipums a good way to look is by looking at the same graph to compare them side by side. We are only interested from the individuals born from 1930 to 1950. We will be doing the same steps in the Question 2 to view p1 and p2. This is the following code :

```
dat1980_clean <- dat1980 %>%
  select(birthyr, educ, age, race, birthqtr, incwage)%>%
  filter(birthyr > 1930 & birthyr < 1950)%>%
  mutate(ln_incwage = ifelse(incwage == 0, 0, log(incwage)))

# Glimpse the filtered data
glimpse(dat1980_clean)
```

```
## Rows: 2,578,544
## Columns: 7
## $ birthyr    <dbl+lbl> 1934, 1948, 1932, 1945, 1941, 1948, 1947, 1937, 1935, 1…
## $ educ       <int+lbl> 6, 6, 6, 6, 6, 11, 11, 11, 8, 6, 10, 9, 6, 8, 7, 6, 8, …
## $ age        <int+lbl> 45, 32, 47, 34, 38, 31, 33, 42, 45, 41, 37, 40, 43, 31,…
## $ race       <int+lbl> 1, 7, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1…
## $ birthqtr   <int+lbl> 4, 1, 3, 3, 4, 2, 1, 2, 1, 2, 1, 1, 3, 1, 1, 4, 4, 3, 1…
## $ incwage    <dbl+lbl>   205,     0, 27605, 20135,     0, 20005, 27005, 30005,…
## $ ln_incwage <dbl> 5.323010, 0.000000, 10.225752, 9.910215, 0.000000, 9.903738…
```

```
educ_mean <- dat1980_clean %>%
  group_by(birthyr, birthqtr) %>%
  summarise(mean_EDUC = mean(educ),
            mean_LWKLYWGE = mean(incwage),
            .groups = 'keep')

#Sort by year of birth
educ_mean <- educ_mean[order(educ_mean$birthyr), ]

#Creation of a variable associating year and quarter of birth
# Updated to include proper YQOB formatting
educ_mean <- educ_mean %>%
  mutate(YQOB = paste(birthyr, birthqtr, sep = "-Q"))  # Update format to YYYY-Q

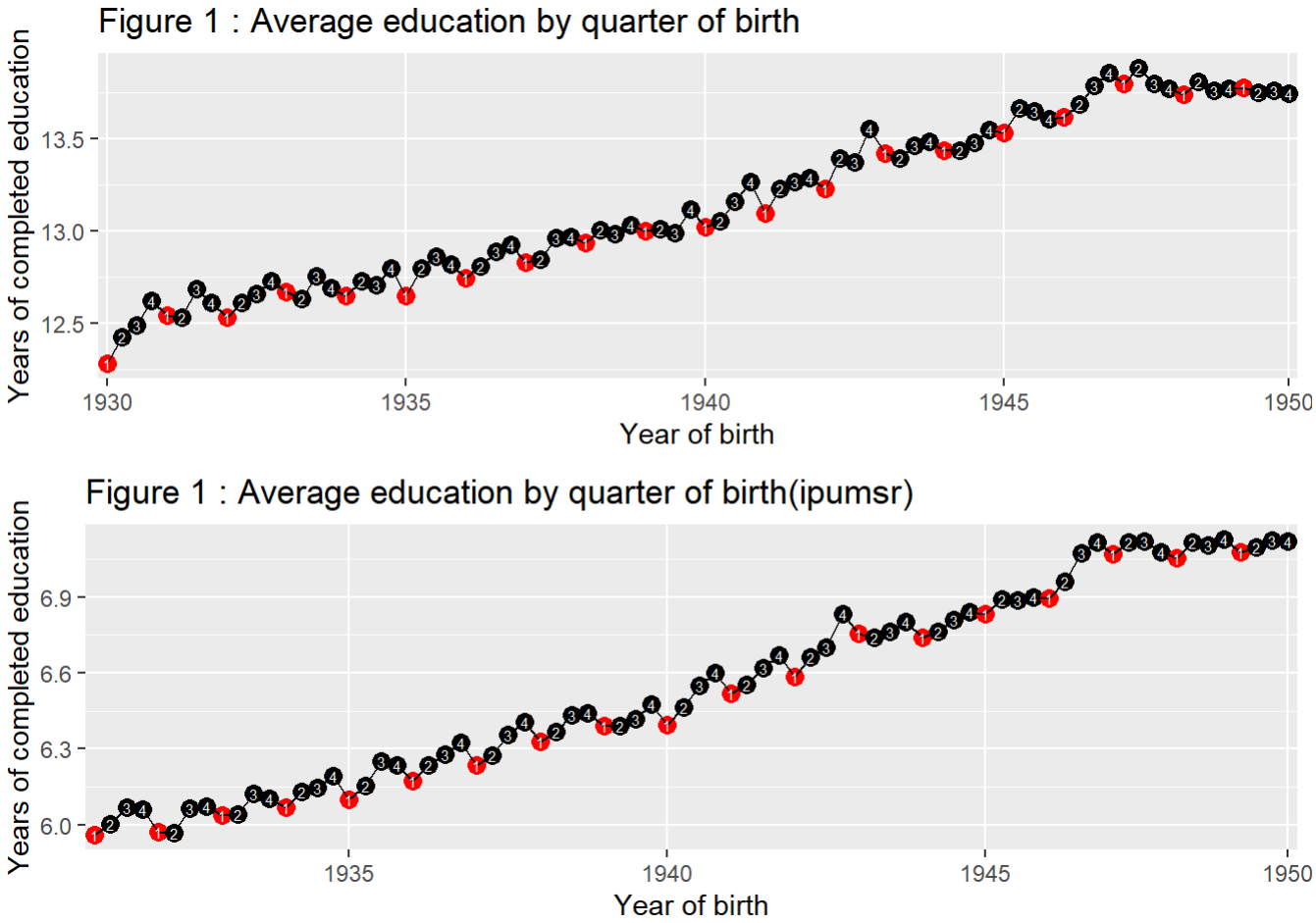
# Updated graphics with corrected x-axis values
p3 <- ggplot(educ_mean, aes(x = YQOB, y = mean_EDUC, label = as.character(birthqtr), color =
as.character(birthqtr))) +
  geom_point(size = 3, shape = 16) +
  geom_line(aes(group = 1), color = "black") +
  geom_text(aes(label = birthqtr), color = "white", size = 2) +
  scale_color_manual(values = c("1" = "red", "2" = "black", "3" = "black", "4" = "black"), gu
ide = "none") +
  labs(x = "Year of birth", y = "Years of completed education", title = "Figure 1 : Average e
ducation by quarter of birth(ipumsr)") +
  scale_x_discrete(breaks = c("1930-Q1", "1935-Q1", "1940-Q1", "1945-Q1", "1949-Q4"),
                   labels = c("1930-Q1" = "1930", "1935-Q1" = "1935", "1940-Q1" = "1940", "19
45-Q1" = "1945", "1949-Q4" = "1950"))

p4 <- ggplot(educ_mean, aes(x = YQOB, y = mean_LWKLYWGE, label = as.character(birthqtr), colo
r = as.character(birthqtr))) +
  geom_point(size = 3, shape = 16) +
  geom_line(aes(group = 1), color = "black") +
  geom_text(aes(label = birthqtr), color = "white", size = 2) +
  scale_color_manual(values = c("1" = "red", "2" = "black", "3" = "black", "4" = "black"), gu
ide = "none") +
  labs(x = "Year of birth", y = "Weekly earnings", title = "Figure 2 : Average weekly wage by
quarter of birth(ipumsr)") +
  scale_x_discrete(breaks = c("1930-Q1", "1935-Q1", "1940-Q1", "1945-Q1", "1949-Q4"),
                   labels = c("1930-Q1" = "1930", "1935-Q1" = "1935", "1940-Q1" = "1940", "19
45-Q1" = "1945", "1949-Q4" = "1950"))
```

IV.1 Difference in the relationship between Year of Birth and

Level of education

```
grid.arrange(p1, p3, nrow = 2)
```



So now that we can see the two graphs together we can see that they have both the same evolution between the two variables. However, we can see that the ipumsUS of 1980 average of the level of education, has more differences between the different years whereas the QQB.raw average doesn't change very much from the different year of birth. This could mean that the data from IPUMS US has more data and more precise.

```
grid.arrange(p2, p4, nrow = 2)
```

Figure 2 : Average weekly wage by quarter of birth

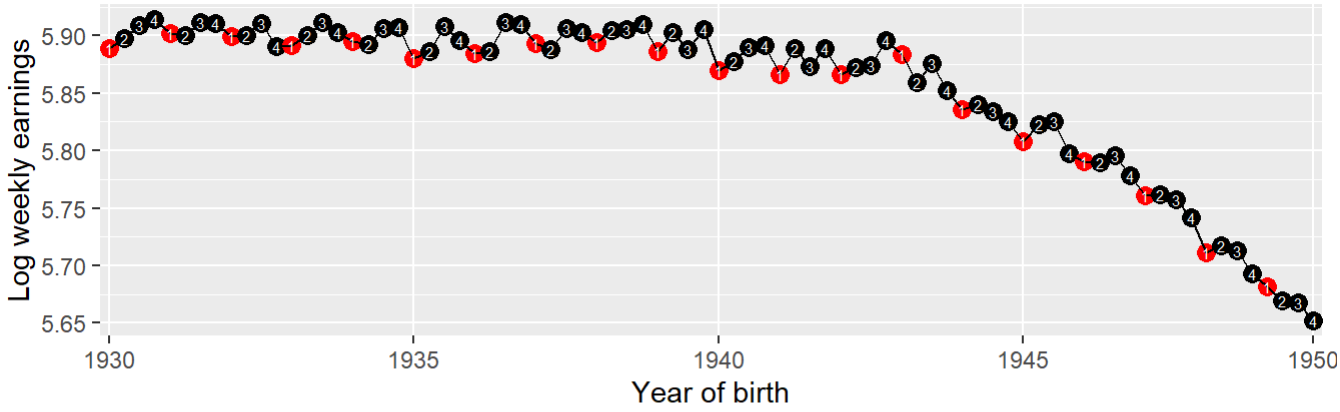
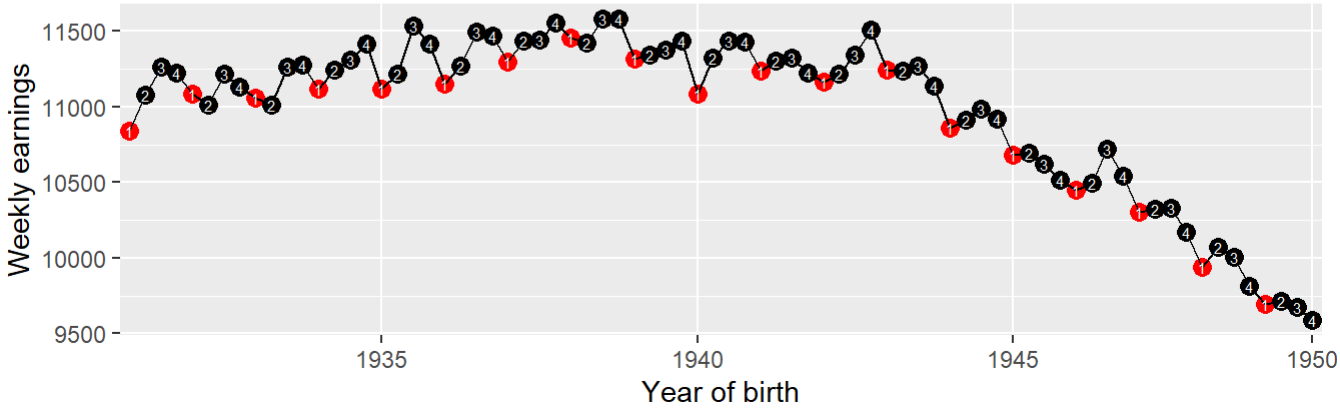


Figure 2 : Average weekly wage by quarter of birth(ipumsr)



Now as we see in the first graph the beginning of the years so from 1930 to 1939 we have a constant correlation with the log of weekly wage income and the year of birth. However in the second graph instead of seeing no growth we see a growth, making us understand that the first decade's average income has been increasing. This could also strengthen the theory in the first contrast between the first figure that the data collected from IPUMS1980 is more accurate.

IV.2 Wald Estimator for Ipums Data

Now to understand the difference between the two data sets we could look at the ceofficents to examine exactly the difference by looking at numbers.

Step 1 : Creating the ln_variables We must make the variable of log of the weekly income wage because otherwise we cannot compare the two data.

```
dat1980_clean <- dat1980 %>%
  select(birthyr, educ, age, race, birthqtr, incwage) %>%
  filter(birthyr > 1930 & birthyr < 1939) %>%
  mutate(
    ln_incwage = ifelse(incwage == 0, 0, log(incwage)), # Handle zero incwage
    wald_dum = as.numeric(birthqtr == 2 | birthqtr == 3 | birthqtr == 4)
  )
```

Step 2 : Following the original code in Question 2 Now that we settled up the correct variable to analyse we can now use the original code found in Question 2.

```r
dat1980_clean$wald_dum <- (dat1980_clean$birthqtr == 3 | dat1980_clean$birthqtr == 4) * 1
# Wald test

# Calculate the means for the groups
mean_ln_wage_q1 <- mean(dat1980_clean$ln_incwage[dat1980_clean$birthqtr == 1], na.rm = TRUE)
mean_ln_wage_q2_4 <- mean(dat1980_clean$ln_incwage[dat1980_clean$wald_dum == 1], na.rm = TRU
E)
mean_education_q1 <- mean(dat1980_clean$educ[dat1980_clean$birthqtr == 1], na.rm = TRUE)
mean_education_q2_4 <- mean(dat1980_clean$educ[dat1980_clean$wald_dum == 1], na.rm = TRUE)

# Calculate the Wald estimator
wald_estimator <- (mean_ln_wage_q1 - mean_ln_wage_q2_4) / (mean_education_q1 - mean_education
_q2_4)

# Calculate the group sizes
n_q1 <- sum(!is.na(dat1980_clean$ln_incwage[dat1980_clean$birthqtr == 1]))
n_q2_4 <- sum(!is.na(dat1980_clean$ln_incwage[dat1980_clean$wald_dum == 1]))

# Calculate the variances
var_ln_wage_q1 <- var(dat1980_clean$ln_incwage[dat1980_clean$birthqtr == 1], na.rm = TRUE)
var_ln_wage_q2_4 <- var(dat1980_clean$ln_incwage[dat1980_clean$wald_dum == 1], na.rm = TRUE)
var_education_q1 <- var(dat1980_clean$educ[dat1980_clean$birthqtr == 1], na.rm = TRUE)
var_education_q2_4 <- var(dat1980_clean$educ[dat1980_clean$wald_dum == 1], na.rm = TRUE)

# Calculate the standard errors for the differences
se_ln_wage_diff <- sqrt(var_ln_wage_q1 / n_q1 + var_ln_wage_q2_4 / n_q2_4)
se_education_diff <- sqrt(var_education_q1 / n_q1 + var_education_q2_4 / n_q2_4)

# Calculate the standard error for the Wald estimator
wald_se <- sqrt(
  (se_ln_wage_diff^2 / (mean_education_q1 - mean_education_q2_4)^2) +
    ((mean_ln_wage_q1 - mean_ln_wage_q2_4)^2 * se_education_diff^2 / (mean_education_q1 - mea
n_education_q2_4)^4)
)


# Run the regression
reg_ols <- lm(ln_incwage ~ educ, data = dat1980_clean)
summary(reg_ols)
```

```
##
## Call:
## lm(formula = ln_incwage ~ educ, data = dat1980_clean)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.301 -5.474  2.031  2.925  6.379
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.845803   0.011042   438.8   <2e-16 ***
## educ        0.314073   0.001642   191.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.112 on 891560 degrees of freedom
## Multiple R-squared:  0.03941,    Adjusted R-squared:  0.0394
## F-statistic: 3.657e+04 on 1 and 891560 DF,  p-value: < 2.2e-16
```

```r
# Adjust the OLS regression for the return to education
reg_ols <- lm(ln_incwage ~ educ, data = dat1980_clean)

# Retrieve the OLS results
ols_return <- coef(reg_ols)["educ"]
ols_se <- summary(reg_ols)$coefficients["educ", 2]

# Create the results table with standard errors
data4 <- data.frame(
  ln_wkly_wage = c(
    round(mean_ln_wage_q1, 5),
    round(mean_ln_wage_q2_4, 5),
    paste0(round(mean_ln_wage_q1 - mean_ln_wage_q2_4, 5), " (", round(se_ln_wage_diff, 5),
")")
  ),
  Education = c(
    round(mean_education_q1, 4),
    round(mean_education_q2_4, 4),
    paste0(round(mean_education_q1 - mean_education_q2_4, 4), " (", round(se_education_diff,
4), ")")
  ),
  Wald_estimator = c(
    "", "",
    paste0(round(wald_estimator, 4), " (", round(wald_se, 4), ")")
  ),
  OLS_return_of_education = c(
    "", "",
    paste0(round(ols_return, 4), " (", round(ols_se, 4), ")")
  ),
  row.names = c(
    "Born in 1st quarter of year",
    "Born in 2nd, 3rd or 4th quarter of year",
    "Difference (std. error) (1)-(2)"
  )
)

# Display with stargazer


# Transpose the data for flipping
data4_flipped <- as.data.frame(t(data4))

# Fix row and column names for better readability
rownames(data4_flipped) <- c(
  "ln (wky. wage)",
  "Education",
  "Wald est. of return to education",
  "OLS return to education"
)
colnames(data4_flipped) <- c("(1)", "(2)", "(3)")

# Generate the flipped table
t2 <- kable(data4_flipped, format = "html", escape = FALSE) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
```

```
  add_header_above(c(" " = 1, "Born in 1st quarter of year" = 1, "Born in 2nd, 3rd, or 4th qu
arter of year" = 1, "Difference (std. error)" = 1))
```

IV.3 Difference between the Wald Estimators

IV.3.1 Wald Estimator Ipums Data

```
t2
```

	Born in 1st quarter of year	Born in 2nd, 3rd, or 4th quarter of year	Difference (std. error)
	1.	2.	3.
ln (wky. wage)	6.72955	6.82778	-0.09823 (0.01088)
Education	6.1114	6.2265	-0.1151 (0.0069)
Wald est. of return to education			0.8536 (0.1073)
OLS return to education			0.3141 (0.0016)

IV.3.2 Wald Estimator pums data

```
t1
```

	Born in 1st quarter of year	Born in 2nd, 3rd, or 4th quarter of year	Difference (std. error)
	1.	2.	3.
ln (wky. wage)	5.8916	5.90269	-0.0111 (0.00274)
Education	12.6881	12.7969	-0.1088 (0.0133)
Wald est. of return to education			0.102 (0.0281)
OLS return to education			0.0709 (3e-04)

By redoing the Wald for the IPUMS table, we see that: Krueger: The Wald estimator is slightly higher than the OLS, which reflects the causal effect of education while controlling for endogeneity and multicolinearity biases. IPUMS: The general intuition remains the same (the Wald estimator is superior to the OLS one, because OLS is downward biased), but the estimated causal effect is much larger. This could indicate that returns to education are higher in IPUMS, but it is necessary to verify whether the instruments and data are comparable. In summary: Intuition is generally maintained (Wald > OLS). However, the significant difference in results in IPUMS prompts examination of assumptions, instruments, and differences between bases.

V.Conclusion

Angrist and Krueger’s (1991) article convincingly demonstrates that education has a positive and causal impact on earnings, using a sound methodological approach based on instrumental variables. Comparison of the results with another datespecialy the waldirms the main idea: education significantly increases income, even if the magnitudes can vary depending on the populations and the instruments used. The methods used complement each other: The Wald estimator provides simple and local intuition, but its scope of application is restricted. TSLS is the most reliable method for estimating an overall causal effect. OLS, although not precise in establishing causality, helps identify bias. By replicating these analyzes with IPUMS, especialy the wald estimator, we validate the robustness of Angrist and Krueger’s findings, while showing that their methodology can be applied in various contexts and on other databases, thus strengthening their relevance for policy analysis educational.